

6.0 INITIALIZATION

Before the ENC28J60 can be used to transmit and receive packets, certain device settings must be initialized. Depending on the application, some configuration options may need to be changed. Normally, these tasks may be accomplished once after Reset and do not need to be changed thereafter.

6.1 Receive Buffer

Before receiving any packets, the receive buffer must be initialized by programming the ERXST and ERXND Pointers. All memory between and including the ERXST and ERXND addresses will be dedicated to the receive hardware. It is recommended that the ERXST Pointer be programmed with an even address.

Applications expecting large amounts of data and frequent packet delivery may wish to allocate most of the memory as the receive buffer. Applications that may need to save older packets or have several packets ready for transmission should allocate less memory.

When programming the ERXST or ERXND Pointer, the internal hardware copy of the ERXWRPT registers will automatically be updated with the value of ERXST. This value will be used as the starting location when the receive hardware begins writing received data. The ERXWRPT registers are updated by the hardware only when a new packet is successfully received.

Note: After writing to ERXST or ERXND, the ERXWRPT registers are not updated immediately; only the internal hardware copy of the ERXWRPT registers is updated. Therefore, comparing if (ERXWRPT == ERXST) is not practical in a firmware initialization routine.

For tracking purposes, the ERXRDPT registers should additionally be programmed with the same value. To program ERXRDPT, the host controller must write to ERXRDPTL first, followed by ERXRDPTH. See **Section 7.2.4 “Freeing Receive Buffer Space”** for more information.

6.2 Transmit Buffer

All memory which is not used by the receive buffer is considered the transmission buffer. Data which is to be transmitted should be written into any unused space. After a packet is transmitted, however, the hardware will write a seven-byte status vector into memory after the last byte in the packet. Therefore, the host controller should leave at least seven bytes between each packet and the beginning of the receive buffer. No explicit action is required to initialize the transmission buffer.

6.3 Receive Filters

The appropriate receive filters should be enabled or disabled by writing to the ERXFCON register. See **Section 8.0 “Receive Filters”** for information on how to configure it.

6.4 Waiting For OST

If the initialization procedure is being executed immediately following a Power-on Reset, the ESTAT.CLKRDY bit should be polled to make certain that enough time has elapsed before proceeding to modify the MAC and PHY registers. For more information on the OST, see **Section 2.2 “Oscillator Start-up Timer”**.

6.5 MAC Initialization Settings

Several of the MAC registers require configuration during initialization. This only needs to be done once; the order of programming is unimportant.

1. Set the MARXEN bit in MACON1 to enable the MAC to receive frames. If using full duplex, most applications should also set TXPAUS and RXPAUS to allow IEEE defined flow control to function.
2. Configure the PADCFG, TXCRCEN and FULDPX bits of MACON3. Most applications should enable automatic padding to at least 60 bytes and always append a valid CRC. For convenience, many applications may wish to set the FRMLNEN bit as well to enable frame length status reporting. The FULDPX bit should be set if the application will be connected to a full-duplex configured remote node; otherwise, it should be left clear.
3. Configure the bits in MACON4. For conformance to the IEEE 802.3 standard, set the DEFER bit.
4. Program the MAMXFL registers with the maximum frame length to be permitted to be received or transmitted. Normal network nodes are designed to handle packets that are 1518 bytes or less.
5. Configure the Back-to-Back Inter-Packet Gap register, MABBIPG. Most applications will program this register with 15h when Full-Duplex mode is used and 12h when Half-Duplex mode is used.
6. Configure the Non-Back-to-Back Inter-Packet Gap register low byte, MAIPGL. Most applications will program this register with 12h.
7. If half duplex is used, the Non-Back-to-Back Inter-Packet Gap register high byte, MAIPGH, should be programmed. Most applications will program this register to 0Ch.
8. If Half-Duplex mode is used, program the Retransmission and Collision Window registers, MACLCON1 and MACLCON2. Most applications will not need to change the default Reset values. If the network is spread over exceptionally long cables, the default value of MACLCON2 may need to be increased.
9. Program the local MAC address into the MAADR1:MAADR6 registers.

REGISTER 6-1: MACON1: MAC CONTROL REGISTER 1

U-0	U-0	U-0	R-0	R/W-0	R/W-0	R/W-0	R/W-0
—	—	—	r	TXPAUS	RXPAUS	PASSALL	MARXEN
bit 7							
							bit 0

Legend:

R = Readable bit

W = Writable bit

U = Unimplemented bit, read as '0'

-n = Value at POR

'1' = Bit is set

'0' = Bit is cleared

x = Bit is unknown

bit 7-5 **Unimplemented:** Read as '0'

bit 4 **Reserved:** Maintain as '0'

bit 3 **TXPAUS:** Pause Control Frame Transmission Enable bit

1 = Allow the MAC to transmit pause control frames (needed for flow control in full duplex)

0 = Disallow pause frame transmissions

bit 2 **RXPAUS:** Pause Control Frame Reception Enable bit

1 = Inhibit transmissions when pause control frames are received (normal operation)

0 = Ignore pause control frames which are received

bit 1 **PASSALL:** Pass All Received Frames Enable bit

1 = Control frames received by the MAC will be written into the receive buffer if not filtered out

0 = Control frames will be discarded after being processed by the MAC (normal operation)

bit 0 **MARXEN:** MAC Receive Enable bit

1 = Enable packets to be received by the MAC

0 = Disable packet reception

REGISTER 6-2: MACON3: MAC CONTROL REGISTER 3

R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0
PADCFG2	PADCFG1	PADCFG0	TXCRCEN	PHDREN	HFRMEN	FRMLNEN	FULDPX
bit 7							bit 0

Legend:

R = Readable bit

W = Writable bit

U = Unimplemented bit, read as '0'

-n = Value at POR

'1' = Bit is set

'0' = Bit is cleared

x = Bit is unknown

bit 7-5

PADCFG2:PADCFG0: Automatic Pad and CRC Configuration bits

111 = All short frames will be zero padded to 64 bytes and a valid CRC will then be appended

110 = No automatic padding of short frames

101 = MAC will automatically detect VLAN Protocol frames which have a 8100h type field and automatically pad to 64 bytes. If the frame is not a VLAN frame, it will be padded to 60 bytes. After padding, a valid CRC will be appended.

100 = No automatic padding of short frames

011 = All short frames will be zero padded to 64 bytes and a valid CRC will then be appended

010 = No automatic padding of short frames

001 = All short frames will be zero padded to 60 bytes and a valid CRC will then be appended

000 = No automatic padding of short frames

bit 4

TXCRCEN: Transmit CRC Enable bit

1 = MAC will append a valid CRC to all frames transmitted regardless of PADCFG bits. TXCRCEN must be set if the PADCFG bits specify that a valid CRC will be appended.

0 = MAC will not append a CRC. The last 4 bytes will be checked and if it is an invalid CRC, it will be reported in the transmit status vector.

bit 3

PHDREN: Proprietary Header Enable bit

1 = Frames presented to the MAC contain a 4-byte proprietary header which will not be used when calculating the CRC

0 = No proprietary header is present. The CRC will cover all data (normal operation).

bit 2

HFRMEN: Huge Frame Enable bit

1 = Frames of any size will be allowed to be transmitted and received

0 = Frames bigger than MAMXFL will be aborted when transmitted or received

bit 1

FRMLNEN: Frame Length Checking Enable bit

1 = The type/length field of transmitted and received frames will be checked. If it represents a length, the frame size will be compared and mismatches will be reported in the transmit/receive status vector.

0 = Frame lengths will not be compared with the type/length field

bit 0

FULDPX: MAC Full-Duplex Enable bit

1 = MAC will operate in Full-Duplex mode. PDPXMD bit must also be set.

0 = MAC will operate in Half-Duplex mode. PDPXMD bit must also be clear.

REGISTER 6-3: MACON4: MAC CONTROL REGISTER 4

U-0	R/W-0	R/W-0	R/W-0	U-0	U-0	R-0	R-0
—	DEFER	BPEN	NOBKOFF	—	—	r	r
bit 7				bit 0			

Legend:

R = Readable bit
-n = Value at POR

W = Writable bit
'1' = Bit is set

U = Unimplemented bit, read as '0'
'0' = Bit is cleared
x = Bit is unknown

- bit 7 **Unimplemented:** Read as '0'
- bit 6 **DEFER:** Defer Transmission Enable bit (applies to half duplex only)
1 = When the medium is occupied, the MAC will wait indefinitely for it to become free when attempting to transmit (use this setting for IEEE 802.3™ compliance)
0 = When the medium is occupied, the MAC will abort the transmission after the excessive deferral limit is reached
- bit 5 **BPEN:** No Backoff During Backpressure Enable bit (applies to half duplex only)
1 = After incidentally causing a collision during backpressure, the MAC will immediately begin retransmitting
0 = After incidentally causing a collision during backpressure, the MAC will delay using the Binary Exponential Backoff algorithm before attempting to retransmit (normal operation)
- bit 4 **NOBKOFF:** No Backoff Enable bit (applies to half duplex only)
1 = After any collision, the MAC will immediately begin retransmitting
0 = After any collision, the MAC will delay using the Binary Exponential Backoff algorithm before attempting to retransmit (normal operation)
- bit 3-2 **Unimplemented:** Read as '0'
- bit 1-0 **Reserved:** Maintain as '0'

REGISTER 6-4: MABBIPG: MAC BACK-TO-BACK INTER-PACKET GAP REGISTER

U-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0
—	BBIPG6	BBIPG5	BBIPG4	BBIPG3	BBIPG2	BBIPG1	BBIPG0
bit 7				bit 0			

Legend:

R = Readable bit
-n = Value at POR

W = Writable bit
'1' = Bit is set

U = Unimplemented bit, read as '0'
'0' = Bit is cleared
x = Bit is unknown

- bit 7 **Unimplemented:** Read as '0'
- bit 6-0 **BBIPG6:BBIPG0:** Back-to-Back Inter-Packet Gap Delay Time bits
When FULDPX (MACON3<0>) = 1:
Nibble time offset delay between the end of one transmission and the beginning of the next in a back-to-back sequence. The register value should be programmed to the desired period in nibble times minus 3. The recommended setting is 15h which represents the minimum IEEE specified Inter-Packet Gap (IPG) of 9.6 μs.
When FULDPX (MACON3<0>) = 0:
Nibble time offset delay between the end of one transmission and the beginning of the next in a back-to-back sequence. The register value should be programmed to the desired period in nibble times minus 6. The recommended setting is 12h which represents the minimum IEEE specified Inter-Packet Gap (IPG) of 9.6 μs.

6.6 PHY Initialization Settings

Depending on the application, bits in three of the PHY module's registers may also require configuration.

The PHCON1.PDPXMD bit partially controls the device's half/full-duplex configuration. Normally, this bit is initialized correctly by the external circuitry (see **Section 2.6 "LED Configuration"**). If the external circuitry is not present or incorrect, however, the host controller must program the bit properly. Alternatively, for an externally configurable system, the PDPXMD bit may be read and the FULDPX bit be programmed to match.

For proper duplex operation, the PHCON1.PDPXMD bit must also match the value of the MACON3.FULDPX bit.

If using half duplex, the host controller may wish to set the PHCON2.HDLDIS bit to prevent automatic loopback of the data which is transmitted.

The PHY register, PHLCON, controls the outputs of LEDA and LEDB. If an application requires a LED configuration other than the default, PHLCON must be altered to match the new requirements. The settings for LED operation are discussed in **Section 2.6 "LED Configuration"**. The PHLCON register is shown in Register 2-2 (page 9).

REGISTER 6-5: PHCON2: PHY CONTROL REGISTER 2

U-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0
—	FRCLNK	TXDIS	r	r	JABBER	r	HDLDIS
bit 15							bit 8

R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0
r	r	r	r	r	r	r	r
bit 7							bit 0

Legend:

R = Readable bit W = Writable bit U = Unimplemented bit, read as '0'
 -n = Value at POR '1' = Bit is set '0' = Bit is cleared x = Bit is unknown

- bit 15 **Unimplemented:** Read as '0'
- bit 14 **FRCLNK:** PHY Force Linkup bit
 1 = Force linkup even when no link partner is detected
 0 = Normal operation
- bit 13 **TXDIS:** Twisted-Pair Transmitter Disable bit
 1 = Disable twisted-pair transmitter
 0 = Normal operation
- bit 12-11 **Reserved:** Write as '0'
- bit 10 **JABBER:** Jabber Correction Disable bit
 1 = Disable jabber correction
 0 = Normal operation
- bit 9 **Reserved:** Write as '0'
- bit 8 **HDLDIS:** PHY Half-Duplex Loopback Disable bit
 When PHCON1<8> = 1 or PHCON1<14> = 1:
 This bit is ignored.
 When PHCON1<8> = 0 and PHCON1<14> = 0:
 1 = Transmitted data will only be sent out on the twisted-pair interface
 0 = Transmitted data will be looped back to the MAC and sent out on the twisted-pair interface
- bit 7-0 **Reserved:** Write as '0'

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NOTES: